

PhD Analysis Examination

January 2023

Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Attempt SIX problems.

- For any two of the following, either give an example or explain why no example exists.
 - a sequence of Lebesgue measurable functions on $\mathbb{R}$ which converges in measure but not Lebesgue a.e.
 - a closed set $E \subset [0, 1]$ with positive Lebesgue measure, but which contains no open intervals
 - a decreasing sequence of nonnegative measurable functions on $\mathbb{R}$ such that $\lim \int f_n \neq \int \lim f_n$
- Prove that for any positive measure μ , the space $L^1(\mu)$ is complete in the L^1 norm.
- Let (μ_n) be a sequence of finite signed measures on a measurable space $(X, \mathcal{M})$. Prove that there exists a finite, positive measure ν on $(X, \mathcal{M})$ such that $\mu_n \ll \nu$ for all n .
- Suppose that $f \in L^1[0, 1]$ and $\int_0^x f \, dm = 0$ for all $x \in [0, 1]$. What can you conclude about f ? (Prove your claim.)
- Let $\mathcal{X}$ be a Banach space. Say that a sequence $(x_n) \subset \mathcal{X}$ converges weakly to $x \in \mathcal{X}$ if $f(x_n) \rightarrow f(x)$ for all $f \in \mathcal{X}^*$. Prove the following:
 - If (x_n) converges weakly, then $\sup_n \|x_n\| < \infty$.
 - Weak limits are unique. (That is, if $x_n \rightarrow x$ weakly and $x_n \rightarrow y$ weakly, then $x = y$.)
- State the closed graph theorem.
 - Let $\mathcal{X}$ be a Banach space with closed subspaces $\mathcal{Y}, \mathcal{Z} \subset \mathcal{X}$. Suppose that every $x \in \mathcal{X}$ admits a *unique* decomposition $x = y + z$ with $y \in \mathcal{Y}, z \in \mathcal{Z}$. Prove that there is a constant C so that $\|y\| \leq C\|x\|$ and $\|z\| \leq C\|x\|$.
- Let H be a Hilbert space and $T : H \rightarrow H$ a bounded linear operator. Define the *adjoint operator* $T^* : H \rightarrow H$ and prove that it is bounded, with $\|T^*\| = \|T\|$.
- (Short answer, you do not need to give a detailed proof, just a brief explanation.)
 - Define the Fourier transform on $L^1(\mathbb{R})$ and sketch its extension to $L^2(\mathbb{R})$.
 - Sketch a construction of a bounded linear functional λ on $L^\infty(\mathbb{R})$ which is *not* of the form $\lambda(f) = \int f(x)g(x) \, dx$ for any L^1 function g .
 - Sketch a proof of the fact that there is no norm under which c_{00} is complete.